How Much Can Conjunction Message Histories Predict Beyond Persistence?

Arjun Vijay Prakash
City Montessori School, Kanpur Road Campus
Lucknow, India

***Abstract*—Can a cutoff-safe history of Conjunction Data Messages predict a later reported $\log_{10} P_c$ better than copying today's report? On the public ESA Collision Avoidance Challenge archive, the mean-error improvement is dominated by later collapses to the reporting floor of -30; it is not a typical-event accuracy gain and not a high-risk decision gain. On a matched cohort of 1459 events, extra MAE beyond persistence falls from 4.49 at 72 h to 0.03 at 12 h, while non-floor ΔMAE is negative at every horizon: off the floor the learned model never beats persistence. Unguarded XGBoost lowers overall MAE from 5.080 to 2.809 on 1659 held-out events, yet floor-excluded MAE rises from 4.073 to 7.562 (Wilcoxon $p = 0.91$). A validation-selected two-part floor hurdle reaches MAE 2.109 with $F_2 = 0$. On ESA's frozen official test, last-risk persistence matches Uriot $L = 0.694$; unguarded XGBoost explodes to $L = 1.75 \times 10^6$ after one overconfident high-risk miss. This is a research measurement, not flight software.**

***Index Terms*—conjunction data message, reported collision probability, forecast horizon, persistence, selective prediction**

I. Introduction

Two catalogued objects can pass close enough to generate a Conjunction Data Message (CDM) [1]. The reported collision probability P_c is recomputed as later tracking shrinks—or fails to shrink—the combined covariance [2]. Waiting usually improves the estimate and consumes planning time. ESA posed the forecasting form of this problem in the 2019 Collision Avoidance Challenge: predict the final reported $\log_{10} P_c$ from messages with time to closest approach of at least two days [3], [4]. Persistence—copying the latest pre-cutoff report—was already strong. Only 12 of 96 teams beat it on ESA-style loss. Conjunction volume has since grown with mega-constellations [5], [6], so the same question is more, not less, relevant: how much extra information sits in an early CDM history?

Physics-based P_c forecasting via expected covariance reduction [7] and operational work on probability dilution [2], [8] suggest that large early covariances can make P_c look safely small, after which later updates either collapse the report to negligible or concentrate it. A recent model predicts which conjunctions will keep high covariance, for extra tracking [9]. The target here is different: the later *reported* $\log_{10} P_c$ under a frozen information cutoff. That target is itself an estimate, not a ground-truth collision occurrence.

This paper makes three contributions. (1) A leakage-safe measurement of extra information beyond persistence as a function of forecast horizon. (2) A decomposition showing that mean-absolute-error (MAE) gains are dominated by collapses to the archive floor -30. (3) Evidence that mean-error optimization can worsen ESA-style high-risk loss. The working hypotheses, and a one-line preview, are: H1 extra information decays toward closest approach (supported on a matched cohort); H2 the MAE win is floor jumps, and off the floor XGBoost is worse than persistence at every matched horizon (supported); H3 extra information does not improve the $\log_{10} P_c \geq -6$ tail under this protocol (supported); H4 covariance volume and the max-risk gap are associated with later movement and floor membership (mixed after partialling).

II. Dataset and Protocol

Labeled CDMs from the ESA training archive comprise 162,634 rows and 13,154 events, 2015–2019, anonymized [4]. An event needs a message at or before the cutoff and a later labeled update, yielding 8,293 eligible events at 48 hours. Train, validation, calibration, and test are event-disjoint (3,731 / 1,659 / 1,244 / 1,659). Seed 42 is the reported local split. Five grouped redraws use seeds 42–46 with no test-tuned knobs. Official-test inputs are public Kelvins `test_data.csv`; labels are the released `true_risk` column from Zenodo 4463683, joined after freeze ($n = 2167$). No hyperparameter was changed after seeing official-test scores. The public file has no calendar dates and no object-pair identifiers, so a year-based split and NORAD grouping are impossible; a mission-grouped split and the official test are the available robustness checks. Fig. 1 records the counts.

Let risk be the latest pre-cutoff reported $\log_{10} P_c$ and y the later reported value. Persistence copies risk. A residual model predicts $\Delta = y - \mathrm{risk}$ and reconstructs $\widehat{y} = \mathrm{risk} + \widehat{\Delta}$. An event is a floor event if $y \leq -30 + 10^{-6}$. That bound is a reporting floor / censored target, not a precise physical P_c. Confirmatory analyses are H1–H4 on the frozen split. Horizon, threshold, and ablation sweeps are exploratory; they are not multiplicity-corrected.

Features are cutoff-safe snapshot fields, history summaries of the same fields, and covariance trends (234 numeric columns after encoding; 77 snapshot, 68 history, 88 covariance trends). The shipped regressor excludes `dilution_gap` $=$ `max_risk_estimate` $-$ `risk`, which is reserved as an H4 probe. XGBoost uses squared error, 250 trees, learning rate 0.05, depth 4, min-child-weight 6, subsample 0.85, column sample 0.8, $\ell_1 = 0.1$, $\ell_2 = 2.0$, seed 42, native NaN handling, and no early stopping. The floor classifier is a shallower

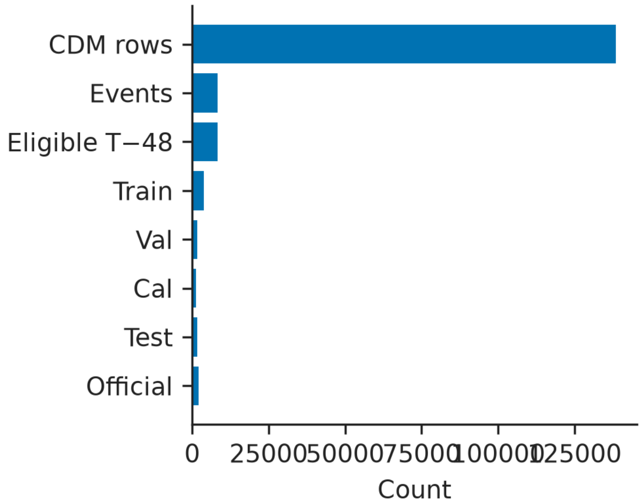

Fig. 1. Archive construction. 162,634 CDM rows yield 8,293 cutoff-safe events at T−48, then an event-disjoint split and a 2167-event official test.

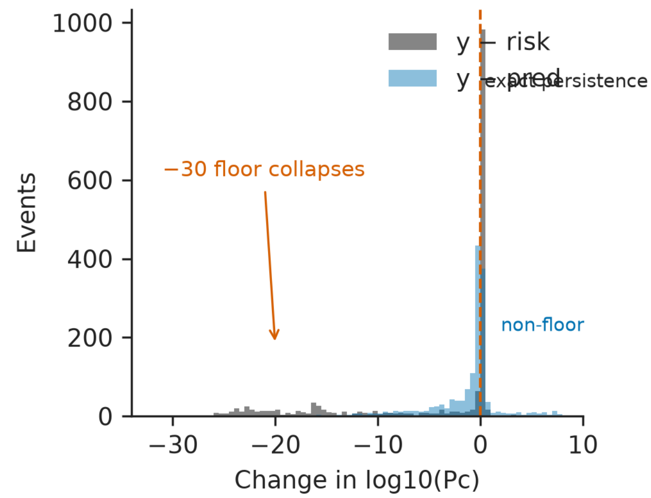

Fig. 2. Later movement $y -$ risk versus unguarded XGBoost residual. The spike is exact persistence; the left tail is collapse to -30.

booster (180 trees, depth 3) with train-set class weighting. Inference is CPU-only.

The live policy is a two-part hurdle: $P(y = -30)$, a residual booster fit on non-floor training events, threshold 0.15 chosen on validation, and no -6 persistence guard. Live intervals are split conformal around that point [10]. SHAP explains the residual regressor. Code: https://github.com/ArjunCodess/PRISM.

III. Structure of the Target

Of 1659 local-test events, 1352 later reports sit at -30 (floor rate 0.815). Persistence already has median absolute error 0: 55.6% of reports do not move after 48 hours. Fig. 2 shows the spike at $y -$ risk $= 0$ and a left tail of floor collapses. Treating the floor as ≤ -20 or ≤ -25 instead of -30 raises the floor rate to 0.845 and 0.828 and still leaves floor-excluded XGBoost MAE worse than persistence. A Tobit-style censored regression is a natural alternative formulation; this study uses a two-part hurdle instead of fitting a Tobit.

Unguarded XGBoost has 48.8% of test events within 0.5 log units and p_{90} absolute error 9.51. Mean error is the wrong headline for this distribution.

IV. Forecasting Experiments

Table I reports the frozen local test. Unguarded XGBoost lowers MAE from 5.080 to 2.809 ($\Delta = 2.270$ $[1.930, 2.638]$) while floor-excluded MAE rises from 4.073 to 7.562. That is H2 on the primary split: mean error improves because later reports hit -30, not because typical events are forecast more accurately. Wilcoxon on paired absolute error does not reject equality ($p = 0.91$). Residual reconstruction is almost the same (MAE 2.760). The floor hurdle reaches MAE 2.109 and median AE 0 because it calls the floor, including on 174 non-floor events; floor-excluded MAE is 9.311. $F_2 = 0$ for

TABLE I

Frozen local test at 48 h ($n = 1659$; 1352 floor). MAE in $\log_{10} P_c$. Δ is MAE(persistence)−MAE(model) with a 95% bootstrap interval.

System	MAE	Med.	Non-fl.	Floor	Δ [95%]	p
Persistence	5.080	0.000	4.073	5.308	—	—
Trend	5.048	—	4.331	5.210	—	—
Unguarded XGB	2.809	0.554	7.562	1.730	2.270 $[1.930, 2.638]$	0.91
Residual XGB	2.760	0.453	7.375	1.712	2.319 $[1.977, 2.685]$	0.92
Floor hurdle	2.109	0.000	9.311	0.474	2.971 $[2.524, 3.415]$	$< 10^{-33}$
Guarded ens.	3.059	0.473	7.332	2.088	2.021 $[1.716, 2.380]$	0.31

the hurdle, residual model, and unguarded XGBoost. ESA-style loss still ties at 0.167 for persistence and an 18 August ensemble snapshot that copies today's report once that report is already ≥ -6.

A clipped one-step trend from the last two cutoff-safe reports barely moves MAE (5.048). On the frozen test, 54.9% of those last-step risk deltas are exactly zero, so a linear extrapolation has almost nothing to project and collapses toward persistence. Snapshot-only XGBoost is 2.911; history-only 2.977; covariance-only 3.148. Adding history summaries to the snapshot lowers MAE from 2.904 to 2.851; covariance trends reach 2.808. Across five grouped redraws, unguarded XGBoost MAE advantage is 2.34 ± 0.04 and floor-excluded MAE is 7.78 ± 0.21. Table II states the pattern without a leaderboard.

The floor classifier at the validation threshold 0.15 has recall 0.976, precision 0.883, AUROC 0.920, AUPRC 0.981, and Brier 0.114 (Fig. 3). The 174 false positives have median later $y = -14.3$; only 25% sit below -20. They are near-floor reports, not high-risk misses. A logistic model on risk alone already reaches AUROC 0.799; adding `max_risk_estimate` or using the gap alone stays near 0.80. XGBoost is better, but the floor phenomenon is visible to a two-feature logistic. Fig. 4 shows why: many later reports

TABLE II
WHAT DOES NOT WORK, RELATIVE TO PERSISTENCE. OFFICIAL L IS URIOT ESA-STYLE LOSS ON THE FROZEN OFFICIAL TEST.

Approach	Mean MAE	Non-floor MAE	Official L
Persistence	baseline	baseline	strong
Trend	no	no	—
Unguarded XGB	yes	no	no
Residual XGB	yes	no	no
Floor hurdle	yes	no	no
Guarded ens.	yes	no	ties

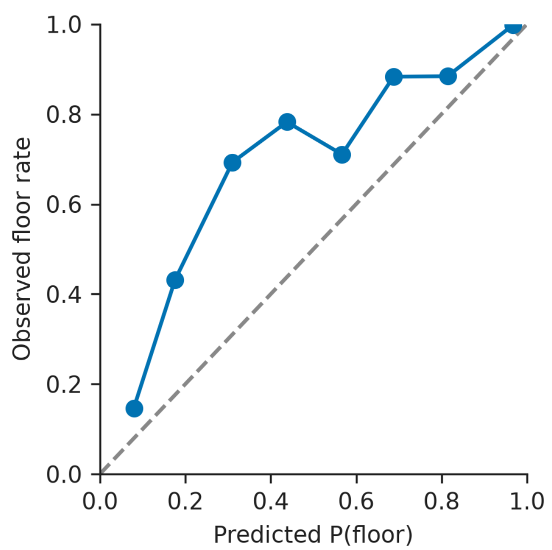

Fig. 3. Reliability of $P(\text{floor})$ for the hurdle classifier.

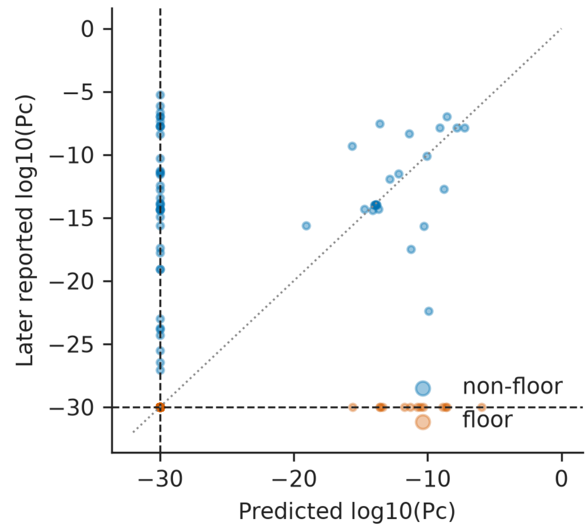

Fig. 4. Floor-hurdle forecast versus later reported $\log_{10} P_c$. The dashed line is the archive floor.

TABLE III
MATCHED-COHORT XGBOOST VERSUS PERSISTENCE. SAME EVENTS AT EVERY HORIZON ($n_{\text{test}} = 1459$).

h	Floor	Overall Δ	[95%]	Non-fl. Δ
72	0.824	4.487	$[3.96, 5.01]$	-3.86
48	0.824	2.198	$[1.76, 2.59]$	-3.95
24	0.824	0.428	$[0.18, 0.65]$	-2.67
12	0.824	0.029	$[-0.15, 0.20]$	-1.70

sit on the -30 line.

V. FORECAST HORIZON

Table III uses the *same* 1459 test events that have valid observations at 72, 48, 24, and 12 hours (7340 events in the archive intersection). Target floor rate is therefore constant (0.824). Extra MAE beyond persistence still falls from 4.49 log units at 72 h to 0.03 at 12 h, where the bootstrap interval covers zero. Non-floor ΔMAE is negative at every horizon (-3.86, -3.95, -2.67, -1.70): off the floor, unguarded XGBoost is worse than persistence at 72, 48, 24, and 12 hours, not only in the pooled 48 h test. Apparent skill is floor-collapse detection, not general forecasting. Fig. 5 shows MAE with those intervals. The older overlapping-set table (advantage 4.53, 2.27, 0.52, 0.06) mixed a changing eligible population with time-to-approach; the matched cohort isolates the horizon effect.

VI. HIGH-RISK EVALUATION

The ESA class is $\log_{10} P_c \geq -6$ [3]. The local test has nine such events. Leave-one-out residual XGBoost on the 66 eligible high-risk events loses to persistence 66/66 (MAE 12.59 versus 1.21). That is clean and statistically fragile. Under this dataset and protocol, these models did not demonstrate improved tail performance. They do not license the broader claim that machine learning cannot forecast high-risk conjunctions.

A `sklearn GroupShuffleSplit` by `mission_id` (seed 42, 20% of *missions* held out) is the available entity-level check. Missions, not events, are the split units; the live floor-hurdle threshold is not retuned, and the same unguarded XGBoost specification is refit on the grouped train split only. The draw puts 15 missions / 3957 events in train and 4 missions / 4336 events in test: event counts are unbalanced because missions differ in volume. The held-out set has 37 high-risk events, so the tail metric is interpretable here: overall XGBoost MAE 3.07 versus persistence 4.74, but high-risk MAE 11.9 versus 1.50. An earlier four-mission draw contained a single high-risk event (MAE 19.2) and is not interpretable for the tail. The public archive cannot test object-pair leakage.

Table IV is confirmation, not discovery. Persistence $L = 0.694$ matches Uriot last-risk-prediction. The selected challenge entry *sesc* reached $L = 0.556$ [3]. Unguarded XGBoost has $F_2 = 0$ and $L = 1.75 \times 10^6$: one wildly overconfident miss on a real high-risk event blows up the numerator while the denominator is zero. The floor hurdle lowers MAE and raises L to 70.5. Fig. 6 annotates that bar. A class-cut sweep on frozen local predictions shows that this study cannot currently say whether the model helps at the tighter operational cuts near -4 to -5; those cells have $n+ \leq 3$ and $F_2 = 0$ for the

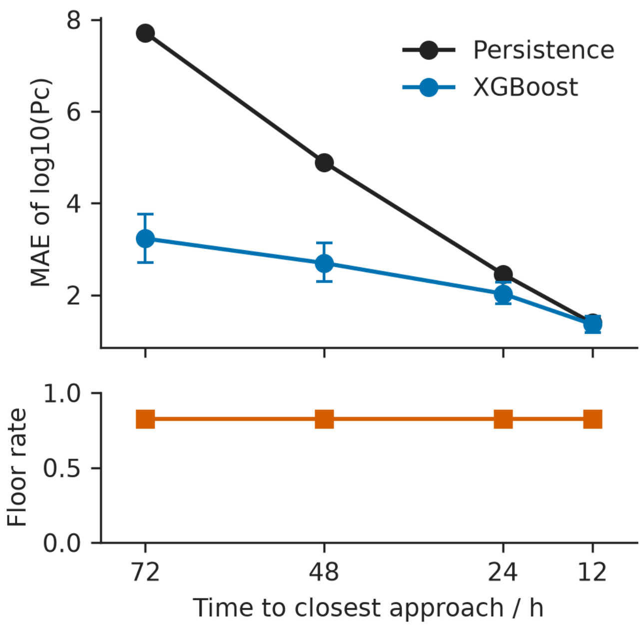

Fig. 5. Matched-cohort MAE versus time to closest approach, with floor rate on the same events.

TABLE IV
OFFICIAL TEST ($n = 2167$; 150 HIGH-RISK; 1673 FLOOR). FROZEN BEFORE LOOK. $L = \mathrm{MSE}_{HR}/F_2$.

System	MAE	Med.	Non-fl.	L	F_2
Persistence (LRP)	5.209	0.000	4.287	0.694	0.739
Unguarded XGB	3.504	0.690	9.333	1.75×10^6	0.000
Residual XGB	3.476	0.594	9.289	104	0.017
Floor hurdle	3.177	0.000	11.832	70.5	0.025
Guarded ens.	3.107	0.439	6.750	0.694	0.739

learned systems.

VII. Covariance and the Max-Risk Gap

Combined snapshot $\log\det$ correlates with later $|y - \text{risk}|$ ($\rho = 0.399$ $[0.36, 0.44]$). The max-risk gap correlates with ending at the floor (logistic test AUC 0.819) and with *less* later movement ($\rho = -0.768$ $[-0.79, -0.75]$). After standardizing and partialling risk, n_{messages}, and miss distance, the $\log\det$ coefficient on $|y - \text{risk}|$ is -0.60: the univariate covariance association is not independently identifiable once today's report is in the model. Association, not cause. Fig. 7 is the scatter; quartile tables remain in the supplement. Residual SHAP is dominated by risk-trajectory fields (`max_risk_scaling`, risk), not covariance volume.

VIII. Uncertainty and Abstention

Bootstrap 90% spread covers 47.7% of the local test. Split conformal around the floor-hurdle point covers 90.1% $[0.887, 0.914]$ with constant width 21.03 log units—wider than the gap from the archive floor to the ESA -6 class. Floor events are well covered (97.6%); non-floor coverage is 57.0%.

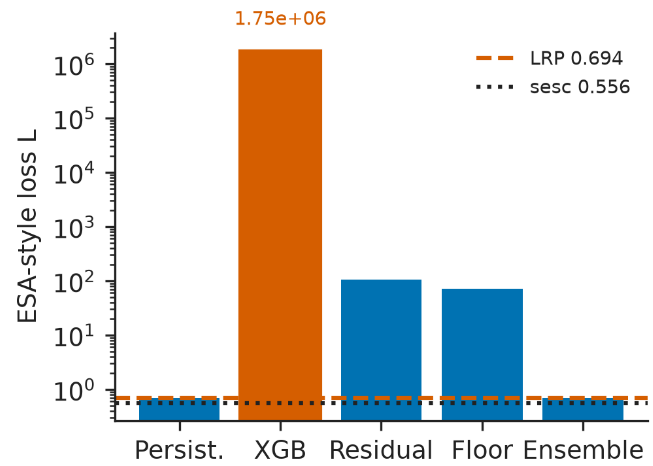

Fig. 6. Official-test ESA-style loss. Persistence matches Uriot LRP 0.694; the selected challenge entry *sesc* is 0.556. The unguarded bar is annotated because it is not a typo.

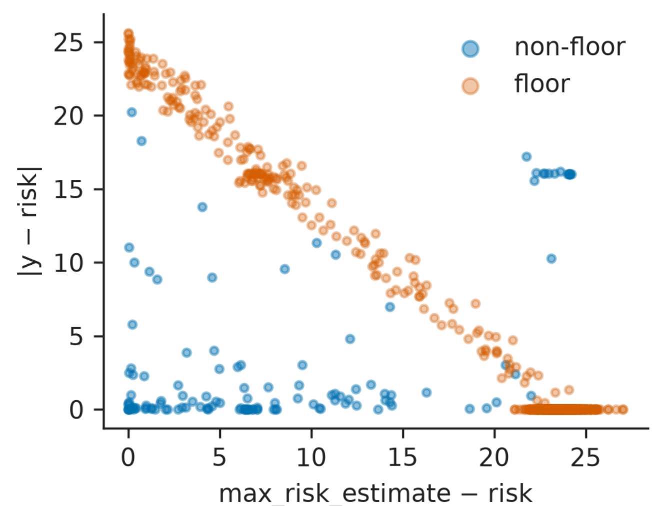

Fig. 7. Max-risk gap versus later $|y - \text{risk}|$. Floor events cluster at large gap and small movement.

The interval is statistically honest and operationally almost useless: a 21-unit band cannot separate a floor collapse from a high-risk update. The 50–80% conformal radii are 0 because most calibration residuals are exact floor hits, so the live rule (abstain when the band crosses -6) does not fire until the 88–90% quantiles. Table V reports accepted-set MAE and false reassurance across that sweep; Fig. 8 plots the same points against nominal coverage rather than acceptance rate, which would collapse the plateau to one marker.

IX. Discussion

Use this model, if at all, to triage reports that are likely to collapse to the archive floor. Do not use it to assess events that are already flagged high-risk. Selecting on MAE ships a floor-calling policy. Selecting on ESA-style loss would have

TABLE V
LIVE CONFORMAL ABSTENTION (BAND CROSSES -6) AT SEVERAL NOMINAL COVERAGES. n_{abs} IS EVENTS SENT TO REVIEW.

Nom.	n_{abs}	Accept	Acc. MAE	FR
50–80%	0	1.00	2.109	9/9
88%	80	0.952	1.812	5/9
90%	159	0.904	1.706	2/9
95%	166	0.900	1.689	2/9
99%	166	0.900	1.689	2/9

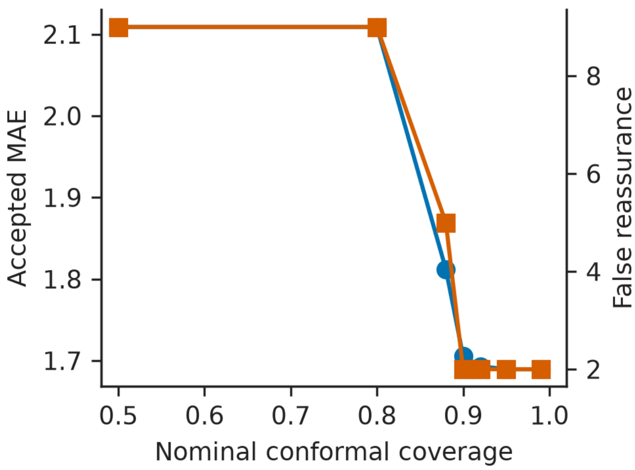

Fig. 8. Accepted MAE and false reassurance versus nominal split-conformal coverage. Radii stay at 0 until 88% because calibration is dominated by exact floor hits.

kept persistence or the guard. Persistence becomes strong near closest approach because later reports have already absorbed later tracking; the matched-cohort decay is the statistical counterpart of that absorption.

The 2015–2019 archive predates today's denser catalogues [5], [6]. Floor-collapse rates could change with tracking cadence and mega-constellation geometry. Manoeuvres are out of scope. The target is later reported $\log_{10} P_c$, not physical collision probability. H1–H4 were pre-specified; sweeps are exploratory. Code, frozen split IDs, hyperparameters, and figure scripts live at https://github.com/ArjunCodess/PRISM; data remain ESA/Zenodo 4463683. This is a research prototype, not a warning system.

ACKNOWLEDGMENT

This work uses the ESA Space Debris Office Collision Avoidance Challenge dataset [3], [4]. ESA, ISRO, NASA, ISTRAC, and CCSDS did not endorse this project.

REFERENCES

[1] Consultative Committee for Space Data Systems, *Conjunction Data Message (CCSDS 508.0-B-1)*, 2013.

[2] M. D. Hejduk and L. C. Johnson, "Satellite conjunction assessment risk analysis for dilution region events: Issues and operational approaches," in *Space Traffic Management Conference*, 2016.

[3] T. Uriot, D. Izzo, L. F. Simões, R. Abay, N. Einecke, S. Rebhan, J. Martinez-Heras, F. Letizia, J. Siminski, and K. Merz, "Spacecraft collision avoidance challenge: Design and results of a machine learning competition," *Astrodynamics*, vol. 6, pp. 121–140, 2022, arXiv:2008.03069.

[4] ESA Advanced Concepts Team and Space Debris Office, "Collision avoidance challenge dataset," Zenodo, 2021.

[5] A. C. Boley and M. Byers, "Satellite mega-constellations create risks in Low Earth Orbit, the atmosphere and on Earth," *Scientific Reports*, vol. 11, p. 10642, 2021.

[6] ESA Space Debris Office, "ESA's annual space environment report," European Space Agency, Tech. Rep. GEN-DB-LOG-00288-OPS-SD, 2024.

[7] M. Duncan, J. Wysack, and J. Frisbee, "Collision probability forecasting using a Monte Carlo simulation," in *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*, 2014.

[8] S. Alfano, "Relating position uncertainty to maximum conjunction probability," in *AAS/AIAA Astrodynamics Specialist Conference*, 2003, AAS 03-548.

[9] J. Stauch, T. Olson, C. Reid, C. Grey, D. C. Kesler, and B. G. Marchand, "Contextual predictive model for early identification of high-covariance conjunctions," *The Journal of the Astronautical Sciences*, vol. 73, p. 19, 2026.

[10] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.